Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Explain the purpose of the diode.

- When the coil switches off it produces a large back emf (voltage spike)
- The diode safely conducts this, protecting the transistor from damage

3 Quick check

- a) Relay
- b) To protect the transistor from reverse voltage
- c) 0.7 V

4 Key terms

- 1 – C
- 2 – B
- 3 – A

5 Explain why a relay is used.

- The transistor circuit can't switch high voltage/current directly
- The relay lets a small current switch a separate high-current circuit safely

6 Explain what is meant by ‘Protection diode’.

- Across the relay coil; protects the transistor from reverse voltage when the coil switches off.